

n-Type Conjugated Polymer Based on a Carbonyl-Terminated Quinoidal Unit for Transparent Conductor

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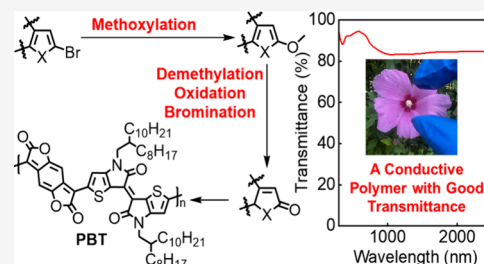


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ABSTRACT: Simultaneously achieving high electrical conductivity and optical transparency remains a formidable challenge for polymers. In this study, we addressed this challenge by preparing a conjugated polymer (PBT) using a carbonyl-terminated quinoidal unit that was synthesized via a newly developed method. PBT is lightly colored because its absorption band is located in the infrared region with minimal absorption in the ultraviolet and visible regions. This unique feature allows infrared photodetection and imaging without sacrificing visible clarity and minimizing interference from visible light. Upon doping, the infrared absorption band in the spectrum of PBT is almost completely suppressed, resulting in broad-spectrum transparency from 300 to 2500 nm with a maximum transmittance of up to 98%. The excellent optical transparency, combined with a close match between the transmission and solar spectrum, ensures that doped PBT has a minimal impact on the color rendering index of light, thereby preserving the color fidelity of objects. Furthermore, the electrical conductivity of doped PBT exceeds 100 S cm^{-1} , the highest reported for intrinsically transparent conjugated polymers. As an active material, doped PBT has been successfully used to fabricate transparent thermoelectric devices. These results highlight the potential of PBT for innovative applications in next-generation transparent electronics.



INTRODUCTION

Transparent conductive materials are unique in that they display both high electrical conductivity and optical transparency, rendering them crucial components in various optoelectronic devices.^{1–6} Transparent conductive polymers, in particular, are highly appealing because of their solution processability and inherent mechanical flexibility, properties that permit their use for flexible, lightweight, and large-area applications.^{1,4} However, the synthesis of transparent conductive polymers is one of the greatest challenges in chemistry and materials science. Vinyl polymers such as polystyrene and poly(methyl methacrylate) exhibit high transparency but are insulators, whereas doped conjugated polymers demonstrate excellent electrical conductivity, but their π -conjugated backbones often result in strong coloration,^{7–11} which compromises their optical transparency. Overcoming the trade-off between conductivity and transparency for polymers is of paramount importance.

Poly(3,4-ethylenedioxythiophene) doped with polystyrene sulfonic acid (PEDOT:PSS) is a widely used transparent conductive p-type (hole-transporting) polymer.^{12–14} In contrast, n-type (electron-transporting) counterparts have rarely been developed. A breakthrough in this area came in the creation of n-PBDF, a self-doped polymer derived from 3,7-dihydrobenzo[1,2-*b*:4,5-*b'*]difuran-2,6-dione.^{15,16} However, both PEDOT:PSS and n-PBDF films absorb strongly in the infrared (IR) region,^{4,15,16} which severely limits their ability to

achieve broad-spectrum transparency. The Boudouris group tackled this limitation by developing a conductive polymer with a nonconjugated backbone bearing side chains with radical sites.¹⁷ This innovation produced a colorless film with excellent optical transmission across a broad spectral range. Nonetheless, the nonconjugated backbone of this polymer impedes charge transport, so the polymer has low electrical conductivity (ca. 0.2 S cm^{-1}).

For achieving broad-spectrum transparency, light-colored polymers are ideal. This can theoretically be achieved by means of two approaches: (1) designing polymers that have an extremely wide bandgap and absorb primarily in the ultraviolet (UV) region or (2) designing polymers that have an extremely narrow bandgap and absorb mainly in the IR region. However, achieving wide-bandgap conjugated polymers is particularly challenging because of their delocalized π -electrons.^{7,18–20} Given this challenge, a more feasible approach to developing lightly colored conjugated polymers involves reducing the bandgap by shifting the absorption band into the IR region. Furthermore, doping can diminish the intensity of the IR

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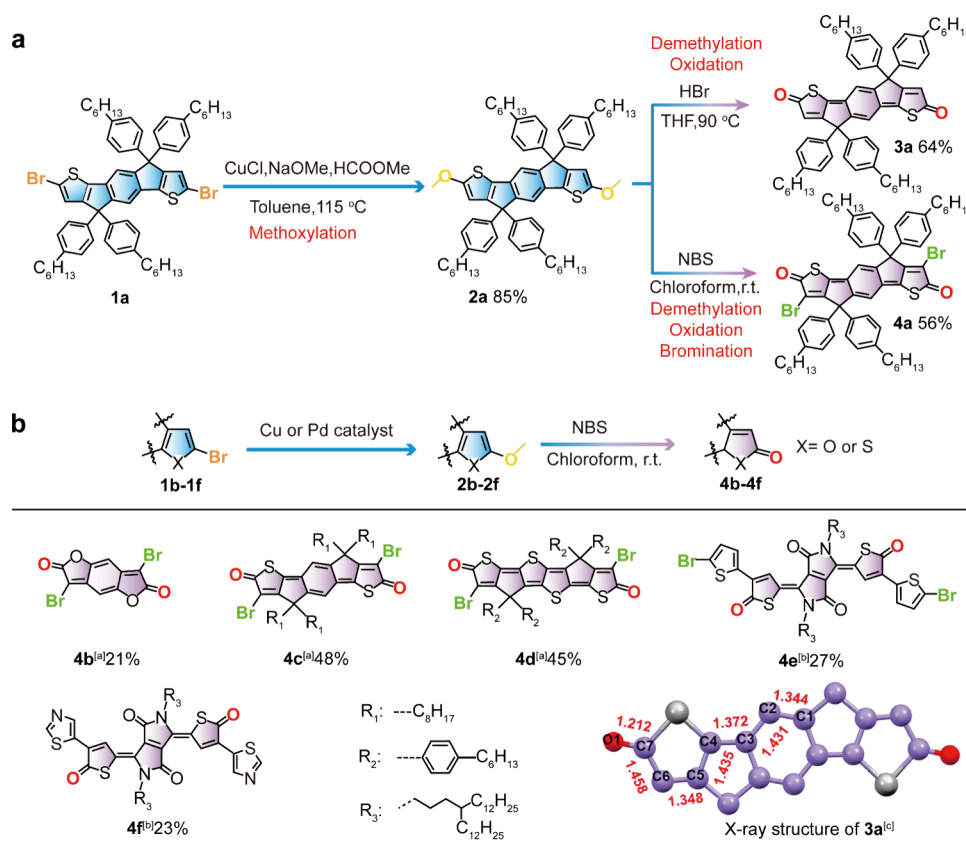


Figure 1. Synthesis of carbonyl-terminated quinoidal compounds. (a) Synthetic routes to **3a** and **4a**. (b) Substrate scope of the method. Yields of isolated products are reported. ^[a]CuCl, NaOMe, HCOOMe, and toluene, 115 °C. ^[b]*t*-BuBrettPhos-Pd-G3, *t*-BuBrettPhos, Cs₂CO₃, toluene, and MeOH, 50 °C. ^[c]X-ray structure of **3a** (CCDC: 2424180) with bond lengths indicated in red.

absorption band, enabling the resulting conductive polymer to exhibit an optical transmission across a broad spectrum.

Herein, we report a doped conjugated polymer (designated PBT) that combines high electrical conductivity with broad-spectrum transparency from the ultraviolet to short-wavelength IR region, which we achieved by using a carbonyl-terminated quinoidal unit as the building block. The quinoidal unit enabled PBT to strongly absorb in the IR region but minimally in the UV and visible regions, resulting in a lightly colored appearance. Such a feature allowed infrared photodetection and imaging without compromising visible clarity and minimizing interference from visible light. Doped PBT not only showed high electrical conductivity (>100 S cm⁻¹) but also demonstrated remarkable transparency across the 300–2500 nm range with a maximum transmittance of 98%. Furthermore, the transmission spectrum of doped PBT matched well with the solar spectrum. Owing to these properties, the doped PBT film exerted a negligible influence on the color rendering index (CRI) of light, achieving a “see-through” character. As a proof of concept, transparent thermoelectric devices using doped PBT as the active layer have been demonstrated.

RESULTS AND DISCUSSION

Synthesis and Characterization of Quinoidal Compounds

Carbonyl-terminated quinoidal structures serve as the core of some natural pigments, such as xylerythrin.²¹ Over the past decade, with rapid advances in organic electronics, interest in such quinoidal compounds has surged, primarily driven by

their electron-accepting properties, planarity,^{22–24} and ability to form polymers that absorb at long wavelengths.^{25–27} The synthetic methods for carbonyl-terminated quinoidal compounds can generally be classified into two categories: (1) multistep routes involving Stille or Suzuki coupling, borylation, oxidation, and bromination^{23,25,28,29} and (2) one-pot bromination–elimination procedures using CuBr₂/NBS.^{27,30} Although the latter approach enables one-pot synthesis, it is applicable only to a limited range of specific substrates. Upon analyzing the processes of the first method, we determined that the dihydroxy compound serves as a key intermediate. Inspired from the straightforward formation of hydroxy compounds from aryl ethers, we hypothesized that carbonyl-terminated quinoidal compounds could be synthesized by methoxylation of aryl halides. To validate this hypothesis, we selected commercially available aryl bromide **1a** as a model substrate (Figure 1a). We found that the methoxylation of **1a** with catalysis by low-cost CuCl smoothly afforded ether **2a** in 85% yield. After purification, **2a** was demethylated using HBr at 90 °C to get the hydroxy intermediate, which was rapidly converted to the target quinoidal compound **3a** in 64% yield during the workup of the reaction. Interestingly, we obtained dibrominated quinoidal compound **4a** (56% yield) directly by treating purified **2a** with NBS by combining demethylation, oxidation, and bromination in a single step, significantly simplifying the synthesis. Notably, this reaction proceeded at room temperature, whereas a temperature of 90 °C was required when HBr was used as the demethylation reagent. Additionally, we found that compound **2a** could be carried forward to the next step without purification, resulting in a

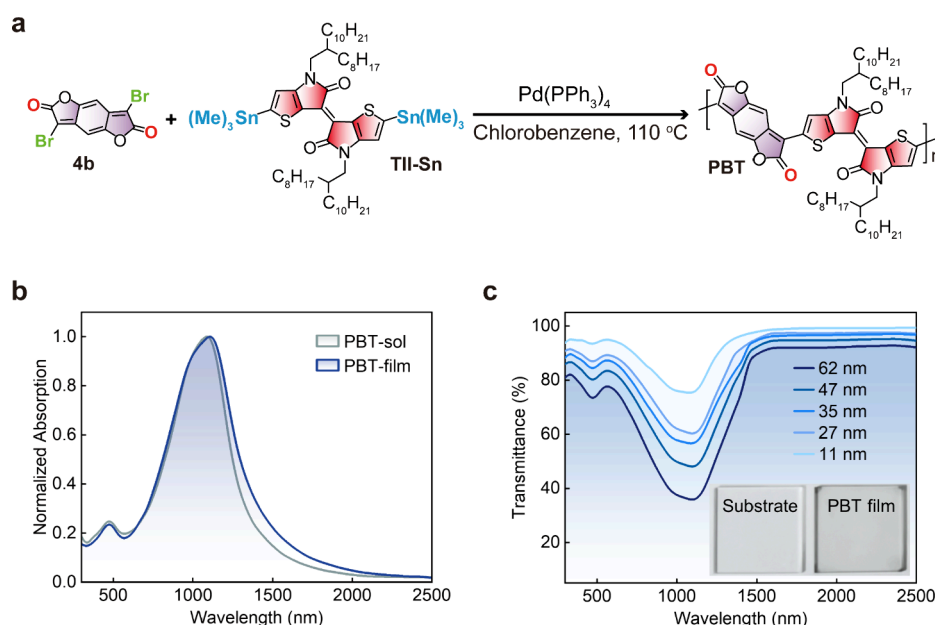


Figure 2. Synthesis and properties of PBT. (a) Synthesis route to PBT. (b) Solution and film (ca. 30 nm) absorption spectra of PBT. Solution absorption spectrum was measured in chlorobenzene with a repeating unit concentration of 10^{-5} mol L⁻¹. Film was prepared by spin-coating the chlorobenzene solution (5 mg mL⁻¹) on quartz substrate. (c) Transmittance spectra of PBT films on quartz substrate with different thicknesses (inset: photographs of a quartz substrate and a PBT film with a thickness of ca. 30 nm). The film thicknesses were controlled by solution concentrations and spin-coating rates (Table S4).

slight improvement in the overall yield (53% yield over two steps). Compared with previously reported protocols for the synthesis of carbonyl-terminated quinoidal compounds,^{23,25} our method offers several advantages: it does not require extra and excessive oxidants and it is operationally straightforward. Moreover, by eliminating the need for *n*-butyllithium to prepare boronic acids, this method can be expected to exhibit good functional group tolerance.

To explore the substrate scope of our new method, we examined polycyclic substrates with different fused rings and obtained desired dibrominated quinoidal compounds **4b–4d** under the optimized conditions (Figure 1b). Additionally, two diketopyrrolopyrrole (DPP)-based substrates with different aromatic substituents were effectively converted to quinoidal compounds **4e** and **4f** (Figure 1b), demonstrating that our method was compatible with amide-containing substrates. However, a palladium catalyst was necessary for the methoxylation of DPP-based substrates,³¹ and bromination of quinoidal compound **4f** did not occur when NBS was used as the reaction reagent. It should be noted that attempts to synthesize compounds **4e** and **4f** via the previously reported oxidation of the diboronic acid method were unsuccessful. The molecular structures of quinoidal compounds **3a** and **4a–4f** were confirmed by using NMR spectroscopy and matrix-assisted time-of-flight mass spectrometry (MALDI-TOF MS) (Figures S1–S21).

The structure of **3a** was further confirmed by X-ray crystallographic analysis. Compound **3a** features a planar conjugated backbone (Figure S22). The C1–C2, C3–C4, and C5–C6 bonds are shorter than a typical C(sp²)–C(sp²) bond (1.45 Å) and are nearly the same length as a typical C(sp²)=C(sp²) bond (1.34 Å). In contrast, the C2–C3, C4–C5, and C6–C7 bonds are longer than the C(sp²)=C(sp²) bond. Furthermore, the length of the C7–O1 bond is about 1.21 Å (Figure 1b), which is consistent with the length of a typical C=O bond (1.22 Å). This bond-length alternation across the

molecule confirmed the quinoidal structure. In the crystal, **3a** adopts a slipped face-to-face π – π stacking arrangement with an interlayer distance of 3.42 Å (Figure S22).

Polymer Design, Synthesis, and Characterization

The brominated quinoidal compound can serve as a versatile building block for the construction of conjugated polymers. Since backbone planarity is critical for efficient intrachain and interchain charge transport,³² density functional theory (DFT) calculations were first performed on the corresponding trimer models to gain insight into the influence of monomer structure on polymer geometry. Thienoisindigo (TII) was selected as the comonomer because of its planar, rigid framework and strong quinoidal character,^{33,34} which are conducive to constructing low bandgap polymers. Furthermore, employing the electron-withdrawing nature of TII to form an acceptor-acceptor polymer backbone is particularly advantageous for achieving *n*-type performance.^{35,36} It should be noted that quinoidal compound **4f** was not considered in the calculations because attempts to brominate **4f** were unsuccessful (Table S1), which precluded subsequent polymer synthesis. The computational results reveal that polymers derived from **4a** and **4d** exhibit severely twisted backbones due to the presence of sterically hindered side chains on the quinoidal monomers. In contrast, compounds **4b**, **4c**, and **4e** are able to form planar polymer backbones with TII, as evidenced by the small dihedral angles (<3°) between the adjacent monomer units in the corresponding trimer models (Figure S23). Accordingly, **4b**, **4c**, and **4e** were selected for polymer synthesis.

The three polymers PBT, PIT-4c, and PDT, derived from **4b**, **4c**, and **4e**, respectively, were synthesized via Stille polycondensation with Pd(PPh₃)₄ as the catalyst and chlorobenzene as the solvent (Figure 2a and Scheme S3). PBT, PIT-4c, and PDT exhibited number-average molecular weight (*M_n*) of 20.3, 29.6, and 31.2 kDa with corresponding polydispersity index (PDI) values of 1.4, 1.5, and 1.4,

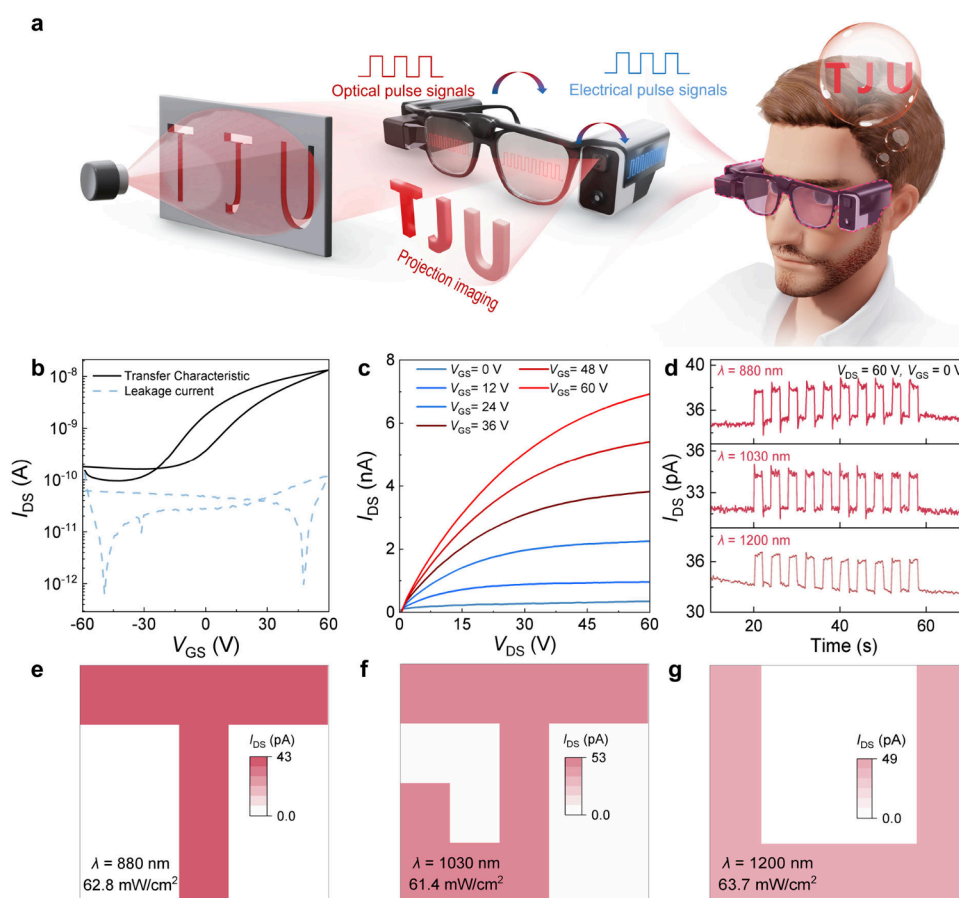


Figure 3. Application of PBT in IR photodetection and imaging. (a) Schematic diagram illustrating the photomask imaging principle: a patterned photomask is illuminated with infrared light, and the transmitted spatial pattern is projected onto a PBT organic thin-film transistor (OTFTs) array, generating corresponding photocurrent changes across the pixelated devices. (b) Transfer characteristics of the OTFTs based on PBT (ca. 25 nm). (c) Output characteristics of the OTFTs based on PBT (ca. 25 nm). The active-layers were deposited by spin-coating from the chlorobenzene solution with a concentration of 4 mg mL⁻¹ and annealed at 150 °C for 10 min. (d) The devices were irradiated with IR light at wavelengths of 880, 1030, and 1200 nm, respectively, under an illumination intensity of about 62 mW/cm². The light was applied in 10 pulses with each pulse lasting 2 s and followed by a 2 s interval. (e–g) Two-dimensional spatial maps of the drain–source current (I_{DS}) distribution across a 5 × 4 OTFT array based on PBT under illumination at (e) λ = 880 nm, (f) λ = 1030 nm, and (g) λ = 1200 nm through shadow masks patterned with the letters “T”, “J”, and “U”, respectively. Upon light irradiation through the shadow mask, the I_{DS} of the illuminated devices increases, which is reflected by the red-colored regions in Figure 3e–g. As a result, the corresponding character patterns can be clearly visualized in the current mapping images.

respectively. The HOMO/LUMO energy levels of PBT, PIT-4c, and PDT, determined by cyclic voltammetry, were −5.44/−4.45, −5.04/−4.07, and −5.02/−4.21 eV, respectively (Figure S24 and Table S2). The experimentally observed trends in both HOMO and LUMO energy levels are consistent with DFT calculations (Figure S23). Among the three polymers, PBT showed the lowest HOMO and LUMO energy levels, which is favorable for realizing n-type charge transport: a low LUMO level stabilizes electron transport, while a deep HOMO level suppresses hole injection.³⁵

All three polymers absorbed strongly in the IR region (Figure 2b and Figure S25) with absorption maxima (λ_{max}) of 1090, 1095, and 1164 nm for PBT, PIT-4c, and PDT, respectively, in dilute chlorobenzene solution. It has been reported that polythienoisindigo (PT) also exhibits strong absorption in the IR region with its absorption spectrum significantly red-shifted relative to those of PBT, PIT-4c, and PDT.³⁷ To elucidate the origin of this difference, DFT calculations were performed for PT and PBT. As shown in Figure S26, incorporation of the quinoidal unit **4b** enhances the quinoidal character of the thienoisindigo moiety in PBT compared to PT; however, the PBT backbone still retains an

alternating aromatic–quinoidal character. In contrast, PT can adopt a fully quinoidal backbone through resonance between aromatic and quinoidal structures. This structural distinction likely accounts for the red-shifted absorption observed for the PT.

Going from solution to thin-film state, the polymers displayed red-shifted absorption spectra with λ_{max} values of 1103, 1104, and 1198 nm for PBT, PIT-4c, and PDT, respectively (Figure 2b and Figure S25). Interestingly, in the thin-film state, PBT exhibited weak absorption in the visible region with the integrated absorbance in the visible range accounting for only ca. 6% of the total integrated absorbance. Consequently, the PBT thin film was lightly colored and highly transparent in the visible range (Figure 2c). At a film thickness of ca. 30 nm, the transmittance in the visible range exceeded 80%. In contrast, PIT-4c and PDT exhibit much stronger absorption contributions in the visible region, resulting in distinctly colored thin films (Figure S25). The unique photophysical properties of PBT make it attractive for the construction of conductors with broad-spectrum transparency. This is further supported by time-dependent density functional theory (TD-DFT) calculations, which indicated that pristine

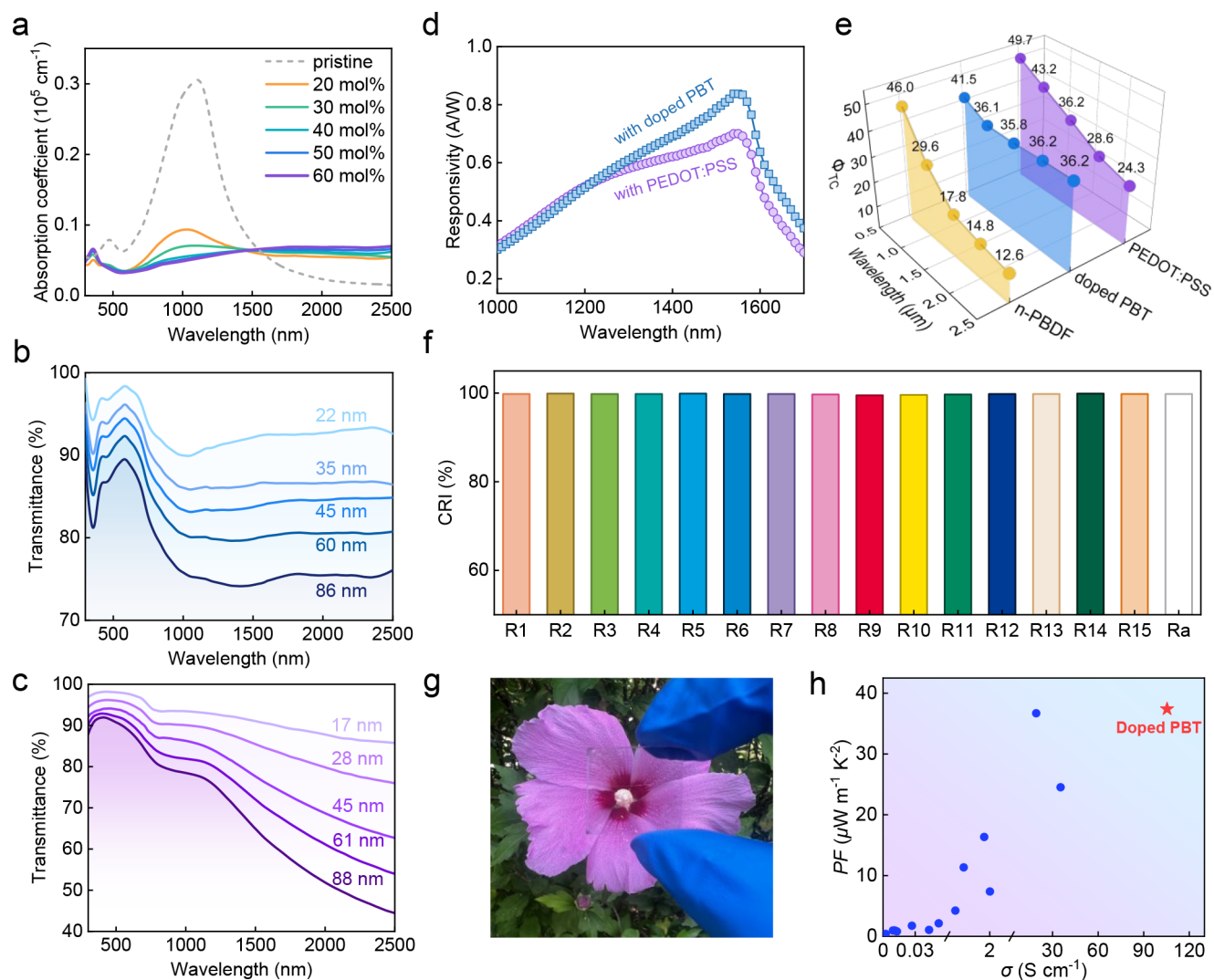


Figure 4. Properties of doped PBT. (a) UV–vis–NIR spectra of PBT films (ca. 45 nm) before and after doping with *N*-DMBI. Films were prepared by spin-coating chlorobenzene solutions (4 mg mL^{-1}) on quartz substrates. (b) Transmittance spectra of 40 mol % doped PBT films (conductivity: ca. 105 S cm^{-1}) on a quartz substrate with different thicknesses. The film thicknesses were controlled by solution concentrations and spin-coating rates (Table S5). (c) Transmittance spectra of PEDOT:PSS (PH1000) films (conductivity: ca. 650 S cm^{-1}) on a quartz substrate with different thicknesses. The film thicknesses were controlled by solution concentrations and spin-coating rates (see Table S6). (d) Responsivity of a commercial Ge detector covered with 40 mol % doped PBT or PEDOT:PSS (PH1000) film (ca. 45 nm). (e) Figures of merit for 40 mol % doped PBT, PEDOT:PSS (PH1000), and n-PBDF at various wavelengths (the unit for Φ_{TC} is $\%(\Omega/\text{sq})^{-1/10}$). (f) CRI of a halogen lamp covered with 40 mol % doped PBT film (ca. 45 nm). (g) Photograph of a flower through 40 mol % doped PBT film (ca. 45 nm). (h) σ and PF values achieved for doped PBT, together with those of n-doped polymers containing a quinoidal unit reported in the literature (please see the data and references in Table S3).

PBT exhibited a strong absorption band in the IR region with a peak at 1039 nm (Figure S27). Upon doping, the absorption band underwent a pronounced red shift, accompanied by a decrease in absorption intensity. At the same time, the absorption contribution in the visible region was reduced after doping. Taken together, the above results demonstrate that PBT is a promising material for an n-type transparent conductor, and it was therefore selected for detailed investigation in this study.

Application of PBT in IR Photodetection and Imaging

The combination of visible transparency and strong IR absorption in PBT enables IR photodetection and imaging, extending human vision into the IR region without compromising the visible clarity and minimizing interference from visible light. Such optoelectronic devices could potentially

be integrated into platforms such as eyeglass substrates (Figure 3a). To evaluate this potential, PBT was employed as the active layer in an organic thin-film transistor (OTFT) array. As expected, PBT displays unipolar n-type transport behavior in OTFTs (Figure 3b,c). In contrast, PIT-4c and PDT display ambipolar transport behavior due to their relatively high HOMO energy levels (ca. -5.0 eV , Table S2), which are insufficient to effectively suppress hole injection and transport.

We first investigated the influence of the semiconductor layer thickness on the photoelectrical properties of PBT-based devices. By controlling the solution concentrations and spin-coating parameters, PBT thin films with three different thicknesses (ca. 25 nm, ca. 50 nm, and ca. 65 nm) were fabricated, and their electrical and photoresponse characteristics were compared. The results show that increasing the film thickness leads to a pronounced increase in the subthreshold

swing (SS) from 49.5 V dec⁻¹ for the ca. 25 nm film to 103.5 V dec⁻¹ for the ca. 50 nm film and to 267 V dec⁻¹ for the ca. 65 nm film (Figure S28). This trend corresponds to a substantial increase in the interfacial defect state density (N_T): 5.18×10^{13} cm⁻² eV⁻¹ (ca. 25 nm), 1.07×10^{14} cm⁻² eV⁻¹ (ca. 50 nm), and 2.80×10^{14} cm⁻² eV⁻¹ (ca. 70 nm). The higher density of defect states in thicker films is detrimental to fast switching, as photogenerated carriers are more readily captured by traps and are released more slowly after the illumination is turned off. Accordingly, the device with a ca. 25 nm-thick active layer exhibited a fast photoresponse with rise and decay times of approximately 150 ms. In contrast, the response time was slowed to about 1050 ms for the ca. 50 nm-thick device and further slowed to approximately 1500 ms for the ca. 65 nm-thick device (Figure S29). Consequently, the following investigations of photodetection and imaging were carried out using devices with an active-layer thickness of ca. 25 nm.

The transfer and output characteristics of the OTFTs with a semiconductor layer thickness of ca. 25 nm are shown in Figure 3b,c. When illuminated with IR LEDs at wavelengths of 880, 1030, and 1200 nm, the devices demonstrated consistent and reliable optoelectronic conversion performance under ten light pulses (Figure 3d). Sensitivity to the IR light intensity was investigated (Figure S30), where the photocurrent is observed to increase proportionally with the light intensity. These results indicate that the devices possess both multipulse stability and light-intensity responsiveness. To further assess imaging performance, a patterned light stimulus using a “T, J, U”-shaped shadow mask was applied. The 5×4 OTFT array displayed distinct current differentiation, and the corresponding photocurrent heatmap clearly reproduced the “T, J, U” pattern at the output terminal (Figure 3e–g), demonstrating the strong potential of PBT-based devices in IR imaging applications.

Properties of Doped PBT

The doping process of PBT was studied by using the widely employed n-type dopant *N,N*-dimethyl-2-phenyl-2,3-dihydro-1*H*-benzimidazole (*N*-DMBI). Upon doping of the polymer, the intensity of the neutral absorption band decreased markedly, accompanied by the emergence of a weak absorption band extending to 2500 nm (Figure 4a). When the dopant ratio was >40 mol %, the neutral absorption band was almost completely quenched, and the doped film showed broad-spectrum transparency from the UV region to the short-wavelength IR region. The optical transmittance of doped PBT thin films as a function of film thickness is shown in Figure 4b. Transmittance spectra were recorded at least three times from various areas of the spin-coated film to ensure accuracy. For a film thickness of ca. 20 nm, the transmittance across the 300–2500 nm range exceeded 90% with a maximum transmittance of 98% at 580 nm on par with the transmittance of PEDOT:PSS (PH1000). Even at a thickness of ca. 85 nm, the doped film maintained more than 75% transmittance over the entire spectral range with a maximum transmittance approaching 90% at 580 nm. Notably, the transmittance of the doped PBT remained nearly constant from 1000 to 2500 nm, whereas the transmittance of PEDOT:PSS (PH1000) films decreased continuously over this range (Figure 4c), as is commonly observed in literature.³⁸ Across all film thicknesses, the transmittance of doped PBT surpassed that of PEDOT:PSS (PH1000) in the 1500–2500 nm range with the difference becoming more pronounced as film thickness

increased. For instance, at a film thickness of ~85 nm, the transmittance of PEDOT:PSS (PH1000) at 1500, 2000, and 2500 nm was 65%, 51%, and 45%, respectively, significantly lower than that of doped PBT at the same wavelengths. Similarly, an ~80 nm film of n-type transparent conductor n-PBDF also showed a sharp drop in transmittance to <40% at wavelengths above 1000 nm and to <20% at 2500 nm,¹⁵ indicating a loss in transparency. Germanium (Ge) detectors covered with a doped PBT film or a PEDOT:PSS (PH1000) film demonstrated comparable responsivity in the 1000–1250 nm range. However, beyond 1250 nm, the PEDOT:PSS (PH1000)-covered detector showed a lower responsivity than the detector covered with doped PBT, with the performance gap widening as the wavelength increased (Figure 4d). This trend is consistent with the progressively decreasing transparency of the PEDOT:PSS (PH1000) film at long wavelengths. These results highlight the good transparency of doped PBT and suggest that it is an excellent candidate for applications demanding broad-spectrum transparent conductors.

CRI is a critical metric for evaluating how accurately and naturally colors are perceived under a specific light source. To assess the effect of a doped PBT film on the CRI, we measured the changes in the CRI after the light passed through the film. The general CRI (R_a) is calculated from the average performance of a light source across 8 standard color samples R1–R8, which represent a range of pastel hues, whereas the special CRI (R_i) is calculated for each of the 15 color samples used by the International Commission on Illumination, providing insights into specific colors, such as saturated colors or skin tones. As shown in Figure 4f, both R_a and R_i of light transmitted through the doped PBT film remained above 99.5, indicating that the colors under the filtered light source appeared virtually identical with those under natural sunlight. Moreover, an image of a flower viewed through the doped PBT film demonstrated excellent color fidelity (Figure 4g). The match between the transmission spectrum and the solar spectrum is another crucial factor in obtaining high CRI values. As shown in Figure S31, the transmission spectrum of doped PBT matched well with the solar spectrum. This good match minimized distortions in the color spectrum, allowing for more natural and accurate color perception under filtered light. Therefore, the excellent performance of doped PBT in preserving natural color rendering can be attributed to its high transparency and the close resemblance between its transmission and the solar spectrum.

The electrical conductivity of the doped PBT thin film (ca. 45 nm) was measured by a four-point-probe method. The optimal doping level was found to be 40 mol %, which yielded an impressive electrical conductivity (σ) of 105.4 S cm⁻¹ (Figure S32). The conductive behavior of doped PBT was further supported by density of states (DOS) analysis (Figure S33), which showed that pristine PBT exhibited a semi-conducting characteristic with a calculated bandgap of 0.56 eV. Upon doping, the Fermi level shifted into the valence band, resulting in a nearly vanishing bandgap and indicating the emergence of conductive behavior. It is recommended that a high-performance transparent conductive film must satisfy the transparency of >80% and an electrical conductivity on the order of 10² S cm⁻¹.^{39,40} For doped PBT, the properties of both high electrical conductivity and broad-spectrum transparency are unique among the polymers reported to date. To our knowledge, the only other reported intrinsic colorless

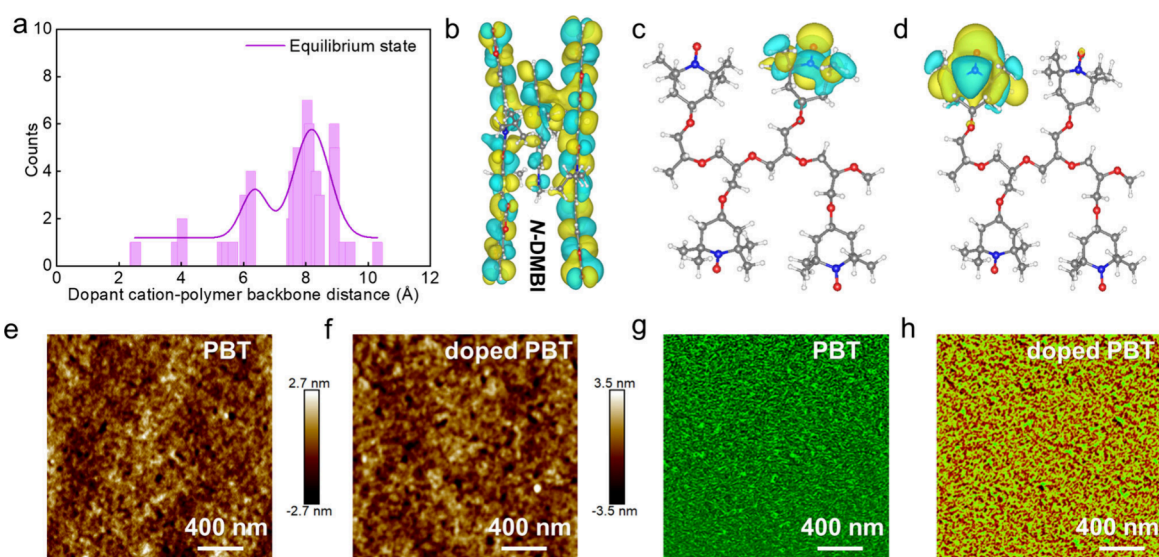


Figure 5. The doping behavior of PBT. (a) Histogram of distances between dopant cations and polymer conjugated backbones for N-DMBI doped PBT. Distance is 0 for polymer backbone, and distance is 12 Å for alkyl chain tails. (b) Orbital overlap between PBT and N-DMBI, obtained from spin-polarized DFT calculations performed using the Vienna Ab initio Simulation Package (VASP). (c) HOMO and (d) LUMO distributions of PTEO, obtained from spin-polarized DFT calculations performed by using VASP. (e, f) AFM height images ($2\ \mu\text{m} \times 2\ \mu\text{m}$) and (g, h) AFM-IR scans ($2\ \mu\text{m} \times 2\ \mu\text{m}$) of (e, g) pristine and (f, h) 40 mol % N-DMBI doped PBT films. The green channel is assigned to PBT at $1,700\ \text{cm}^{-1}$, while the red channel is assigned to N-DMBI at $1,552\ \text{cm}^{-1}$.

conductive polymer (PTEO) was achieved by incorporating radical centers onto the side chains of a nonconjugated backbone. However, this approach resulted in an electrical conductivity of only ca. $0.2\ \text{S cm}^{-1}$,¹⁷ which is 3 orders of magnitude lower than that of doped PBT.

Figure S34 presents the Arrhenius fits for the temperature-dependent conductivities of various doped PBT films over the temperature range of 123–353 K. As the doping level increases, the activation energy decreases, indicating that the Fermi level shifts upward toward the transport energy level.⁴¹ Besides, the conductivity increased with increasing temperature over the entire measured range, suggesting that charge transport in doped PBT is governed by a thermally activated hopping mechanism.^{42,43} This behavior contrasts with that of nonconjugated PTEO, which exhibits Arrhenius-type temperature dependence only within a limited high-temperature range.¹⁷

The air stability of doped PBT was assessed by monitoring the electrical conductivity of doped films under ambient conditions (relative humidity = $\sim 30\%$) without encapsulation (Figure S35). The conductivity decreased by ca. 10% within the first day of air exposure and retained ca. 70% of its initial value after 25 days. This performance exceeds that of most reported polymeric materials, whose n-doped states typically undergo rapid degradation under ambient conditions^{44–46} with only a few examples demonstrating air stability beyond 10 days.^{15,16,47,48} The relatively good air stability of doped PBT can be attributed to its low-lying LUMO energy level ($-4.45\ \text{eV}$, one of the lowest reported for n-type polymers) as well as the reduction-induced aromatic stabilization of the quinoidal unit upon n-doping.^{49,50}

A figure of merit (Φ_{TC}) is a simple yet effective parameter to assess the quality of transparent conductive materials.⁵¹ It is calculated as $T/\sqrt{10}R_s$, where T represents the optical transmittance (0–100%) of a film and R_s corresponds to its electrical sheet resistance (Ω/sq). From the formula, it is

evident that T takes a more value on Φ_{TC} than R_s . We calculated the Φ_{TC} values for PEDOT:PSS (PH1000), doped PBT, and n-PBDF films at 550, 1000, 1500, 2000, and 2500 nm. The thicknesses of doped PBT, PEDOT:PSS (PH1000), and n-PBDF films were fixed to be similar (ca. 85 nm). Under these conditions, the corresponding sheet resistances were 1650, 191, and $21\ \Omega/\text{sq}$. At 550 nm, all three films exhibited Φ_{TC} values of >40 (Figure 4e). However, as the wavelength increased, the Φ_{TC} values for PEDOT:PSS (PH1000) and n-PBDF continuously declined. In particular, the Φ_{TC} values of PEDOT:PSS (PH1000) and n-PBDF films decreased sharply to 24.3 and 12.6, respectively, at 2500 nm. In contrast, the Φ_{TC} of doped PBT remained nearly constant at ca. 36 at 1000, 1500, 2000, and 2500 nm, a finding that is consistent with its stable optical transmittance in the 1000–2500 nm region. The Φ_{TC} values of doped PBT in the short-wavelength IR region rivaled those of PEDOT:PSS (PH1000) and were significantly higher than those of n-PBDF, reaffirming the excellent performance of doped PBT as a transparent, conductive material.

Thermoelectric devices can directly convert waste heat into electricity^{46,52–54} and are a promising green energy solution. To assess the thermoelectric performance of doped PBT, the Seebeck coefficient (S) was obtained by imposing a temperature difference across a sample (Figure S36). Doped PBT showed an S of $-59.6\ \mu\text{V K}^{-1}$. The negative S value confirms the n-type transport behavior of PBT. The power factor (PF) of the thermoelectric device was calculated to be $37.4\ \mu\text{W m}^{-1}\ \text{K}^{-2}$, according to the equation of $\text{PF} = S^2\sigma$. To the best of our knowledge, this PF is among the highest reported to date for n-doped quinoid-based π -conjugated polymers (Figure 4h and Table S3).⁵⁵ Notably, the thermoelectric device also showed good transparency (Figure S37).

Theoretical calculations were conducted to elucidate why doped PBT exhibits a significantly higher conductivity than PTEO, which was reported by the Boudouris group. Molecular dynamics simulations were first used to predict the positions of

N-DMBI⁺ cations within the polymer matrix. The optimized structures were placed in a supercell under periodic boundary conditions. At equilibrium, most *N*-DMBI⁺ cations were located away from the polymer backbones, residing instead in the alkyl side-chain packing regions to form homogeneous doping (Figure 5a). However, the small fraction of *N*-DMBI molecules located near the backbone can effectively dope the polymer, as indicated by the obvious orbital coupling between *N*-DMBI and PBT (Figure 5b).

If most of the dopant was localized on the polymer backbone, the molecular packing would be significantly disrupted, particularly the π - π stacking distance. To further support the homogeneous doping behavior of PBT, 2D grazing incidence wide-angle X-ray scattering (2D-GIWAXS) measurements were performed on both pristine and doped films. Analysis of the GIWAXS patterns revealed that the microstructure of PBT remained nearly unchanged with similar lamellar packing distances (~ 21.8 Å) and π - π stacking distances (~ 3.5 Å) before and after doping (Figure S38).

Additional evidence for homogeneous doping is the suppression of large-scale phase separation, resulting in a smooth and uniform film morphology. Atomic force microscopy (AFM) images revealed similar morphologies for pristine and doped films with no obvious aggregates. Comparable results were obtained from AFM coupled with infrared spectroscopy measurements (AFM-IR; Figure 5e-h), which provides chemical analysis with nanoscale spatial resolution by using an AFM tip to locally detect the thermal expansion of a sample induced by IR absorption.⁵⁶ By imaging at characteristic IR wavelengths corresponding to the absorption peaks of PBT (1700 cm^{-1}) and *N*-DMBI (1552 cm^{-1}) (Figure S39), the spatial distribution of the polymer and dopant could be visualized. The resulting images clearly demonstrate that PBT and *N*-DMBI are well mixed, forming a homogeneously doped film. Besides, as mapped by AFM-IR, the pristine PBT formed fibrillar domains with an average diameter of 112 nm, which decreased slightly to 110 nm in the doped films.

The unchanged microstructure after doping is advantageous for charge transport.⁵⁷ Additionally, the polaron in the PBT anion exhibited good delocalization along the polymer chain (Figure S40), which facilitates charge carrier hopping between the polymer backbones. In contrast, the frontier orbitals of PTEO are mainly localized at the radical sites (Figure 5c,d). As a result, its conductivity arises solely from the formation of a percolating network of open-shell sites, making it dependent on the channel length between the electrodes. Consequently, the combination of homogeneous doping and efficient polaron delocalization accounts for the high electrical conductivity of PBT.

CONCLUSION

In this study, we synthesized a lightly colored conjugated polymer (designated PBT) based on a carbonyl-terminated quinoidal unit, which was prepared by a newly developed method involving methoxylation, followed by one-pot demethylation, oxidation, and bromination. The lightly colored nature of PBT is derived from its strong IR absorption with minimal absorption in the ultraviolet and visible regions, enabling IR photodetection and imaging without compromising visible clarity and minimizing interference from visible light. This unique feature also allowed the realization of doped PBT with broad-spectrum transparency ranging from 300 to

2500 nm with a maximum transmittance of 98%. Even at a film thickness of ca. 85 nm, doped PBT maintained more than 75% transmittance from 300 to 2500 nm with a maximum transmittance of approaching 90%. Furthermore, the transmission spectrum of doped PBT matched well with the solar spectrum. Owing to these attributes, doped PBT showed a near-perfect CRI, effectively preserving the natural appearance of colors. In addition, doped PBT exhibited high electrical conductivity of up to 105.4 S cm^{-1} , 3 orders of magnitude higher than that of previously reported intrinsically transparent conductive polymers. As a proof of concept, transparent thermoelectric devices using doped PBT as the active layer have been demonstrated. The combination of high conductivity with broad-spectrum transparency in doped PBT has not been observed in previously reported conjugated polymers. Our findings offer new insights into the puzzling relationship between electrical conductivity and optical transparency in polymers, opening avenues for the development of advanced conductive materials for transparent organic electronics.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c20042>.

Synthesis and characterization of quinoidal compounds and polymer, including the NMR and MALDI-TOF spectra, TGA, DSC, and CV curves, device fabrication and characterization, and theory calculations (PDF)

Accession Codes

Deposition Number 2424180 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

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REFERENCES

- (1) Guo, X.; Facchetti, A. The journey of conducting polymers from discovery to application. *Nat. Mater.* **2020**, *19* (9), 922–928.
- (2) Ellmer, K. Past achievements and future challenges in the development of optically transparent electrodes. *Nat. Photonics* **2012**, *6* (12), 809–817.
- (3) Yu, X.; Marks, T. J.; Facchetti, A. Metal oxides for optoelectronic applications. *Nat. Mater.* **2016**, *15* (4), 383–396.
- (4) Aivali, S.; Beaumont, C.; Leclerc, M. Conducting polymers: towards printable transparent electrodes. *Prog. Polym. Sci.* **2024**, *148*, 101766.
- (5) Chen, Z.; Wang, J.; Wu, H.; Yang, J.; Wang, Y.; Zhang, J.; Bao, Q.; Wang, M.; Ma, Z.; Tress, W.; Tang, Z. A transparent electrode based on solution-processed ZnO for organic optoelectronic devices. *Nat. Commun.* **2022**, *13* (1), 4387.
- (6) Hecht, D. S.; Hu, L.; Irvin, G. Emerging transparent electrodes based on thin films of carbon nanotubes, graphene, and metallic nanostructures. *Adv. Mater.* **2011**, *23* (13), 1482–1513.
- (7) Conjugated aromatic systems. *Nat. Synth.* **2023**, *2* (9), 799–799.
- (8) Morita, F.; Kishida, Y.; Sato, Y.; Sugiyama, H.; Abekura, M.; Nogami, J.; Toriumi, N.; Nagashima, Y.; Kinoshita, T.; Fukuhara, G.; Uchiyama, M.; Uekusa, H.; Tanaka, K. Design and enantioselective synthesis of 3D π -extended carbohelicenes for circularly polarized luminescence. *Nat. Synth.* **2024**, *3* (6), 774–786.
- (9) Xiong, H.; Lin, Q.; Lu, Y.; Zheng, D.; Li, Y.; Wang, S.; Xie, W.; Li, C.; Zhang, X.; Lin, Y.; Wang, Z.-X.; Shi, Q.; Marks, T. J.; Huang, H. General room-temperature Suzuki-Miyaura polymerization for organic electronics. *Nat. Mater.* **2024**, *23* (5), 695–702.
- (10) Yang, K.; Li, Z.; Huang, Y.; Zeng, Z. bay/ortho-Octa-substituted Perylene: A Versatile Building Block toward Novel Polycyclic (Hetero)Aromatic Hydrocarbons. *Acc. Chem. Res.* **2024**, *57* (5), 763–775.
- (11) Luo, T.; Wang, Y.; Hao, J.; Chen, P.-A.; Hu, Y.; Chen, B.; Zhang, J.; Yang, K.; Zeng, Z. Furan-Extended Helical Rylenes with Fjord Edge Topology and Tunable Optoelectronic Properties. *Angew. Chem., Int. Ed.* **2023**, *62* (5), No. e202214653.
- (12) Gueye, M. N.; Carella, A.; Faure-Vincent, J.; Demadrille, R.; Simonato, J.-P. Progress in understanding structure and transport properties of PEDOT-based materials: A critical review. *Prog. Mater. Sci.* **2020**, *108*, 100616.
- (13) Yano, H.; Kudo, K.; Marumo, K.; Okuzaki, H. Fully soluble self-doped poly(3,4-ethylenedioxythiophene) with an electrical conductivity greater than 1000 S cm⁻¹. *Sci. Adv.* **2019**, *5* (4), No. eaav9492.
- (14) Elschner, A.; Kirchmeyer, S.; Lovenich, W.; Merker, U.; Reuter, K. *PEDOT: principles and applications of an intrinsically conductive polymer*; CRC Press, 2010.
- (15) Ke, Z.; Abtahi, A.; Hwang, J.; Chen, K.; Chaudhary, J.; Song, I.; Perera, K.; You, L.; Baustert, K. N.; Graham, K. R.; Mei, J. Highly conductive and solution-processable n-doped transparent organic conductor. *J. Am. Chem. Soc.* **2023**, *145* (6), 3706–3715.
- (16) Tang, H.; Liang, Y.; Liu, C.; Hu, Z.; Deng, Y.; Guo, H.; Yu, Z.; Song, A.; Zhao, H.; Zhao, D.; Zhang, Y.; Guo, X.; Pei, J.; Ma, Y.; Cao, Y.; Huang, F. A solution-processed n-type conducting polymer with ultrahigh conductivity. *Nature* **2022**, *611* (7935), 271–277.
- (17) Joo, Y.; Agarkar, V.; Sung, S. H.; Savoie, B. M.; Boudouris, B. W. A nonconjugated radical polymer glass with high electrical conductivity. *Science* **2018**, *359* (6382), 1391–1395.
- (18) Ding, L.; Yu, Z.-D.; Wang, X.-Y.; Yao, Z.-F.; Lu, Y.; Yang, C.-Y.; Wang, J.-Y.; Pei, J. Polymer semiconductors: synthesis, processing, and applications. *Chem. Rev.* **2023**, *123* (12), 7421–7497.
- (19) Zhao, R.; Liu, J.; Wang, L. Polymer acceptors containing B←N units for organic photovoltaics. *Acc. Chem. Res.* **2020**, *53* (8), 1557–1567.
- (20) Zhou, Z.; Yang, K.; He, L.; Wang, W.; Lai, W.; Yang, Y.; Dong, Y.; Xie, S.; Yuan, L.; Zeng, Z. Sulfone-Functionalized Chichibabin's Hydrocarbons: Stable Diradicaloids with Symmetry Breaking Charge Transfer Contributing to NIR Emission beyond 900 nm. *J. Am. Chem. Soc.* **2024**, *146* (10), 6763–6772.
- (21) Carey, J. L.; Shand, C. A.; Thomson, R. H.; Greenhalgh, C. W. Naphthoquinone mono- and di-methide lactones. *J. Chem. Soc., Perkin Trans. 1* **1984**, No. 0, 1957–1962.
- (22) Osaka, I.; Abe, T.; Mori, H.; Saito, M.; Takemura, N.; Koganezawa, T.; Takimiya, K. Small band gap polymers incorporating a strong acceptor, thieno[3,2-b]thiophene-2,5-dione, with p-channel and ambipolar charge transport characteristics. *J. Mater. Chem. C* **2014**, *2* (13), 2307–2312.
- (23) Kawabata, K.; Osaka, I.; Nakano, M.; Takemura, N.; Koganezawa, T.; Takimiya, K. Thienothiophene-2,5-dione-based donor-acceptor polymers: improved synthesis and influence of the donor units on ambipolar charge transport properties. *Adv. Electron. Mater.* **2015**, *1* (6), 1500039.

- (24) Deng, Y.-f.; Wang, T.-z.; Geng, X.-k. Recent progress of conjugated polymers containing quinoidal structure. *Acta Polym. Sin.* **2023**, *54* (6), 803–817.
- (25) Kawabata, K.; Saito, M.; Osaka, I.; Takimiya, K. Very small bandgap π -conjugated polymers with extended thienoquinoids. *J. Am. Chem. Soc.* **2016**, *138* (24), 7725–7732.
- (26) Mikie, T.; Osaka, I. Small-bandgap quinoid-based π -conjugated polymers. *J. Mater. Chem. C* **2020**, *8* (41), 14262–14288.
- (27) Cai, H.; Tang, H.; Wang, T.; Xu, C.; Xie, J.; Fu, M.; Luo, X.; Hu, Z.; Zhang, Y.; Deng, Y.; Li, G.; Liu, C.; Huang, F.; Cao, Y. An n-type open-shell conjugated polymer with high-spin ground-state and high intrinsic electrical conductivity. *Angew. Chem., Int. Ed.* **2024**, *63* (25), No. e202402375.
- (28) Testaferri, L.; Tiecco, M.; Zanirato, P.; Martelli, G. Preparation and tautomeric structures of some potential 2,5-dihydroxythieno[3,2-*b*]thiophenes. *J. Org. Chem.* **1978**, *43* (11), 2197–2200.
- (29) Chen, S.; Bolag, A.; Nishida, J.-i.; Yamashita, Y. n-Type field-effect transistors based on thieno[3,2-*b*]thiophene-2,5-dione and the bis(dicyanomethylene) derivatives. *Chem. Lett.* **2011**, *40* (9), 998–1000.
- (30) Nakatsuka, M.; Nakasuji, K.; Murata, I.; Watanabe, I.; Saito, G.; Enoki, T.; Inokuchi, H. 3,7-DIHALO-2H,6H-BENZO[1,2-*b*:4,5-*b'*]DITHIOPHENE-2,6-DIONE. NEW WURSTER-TYPE ACCEPTORS ISOELECTRONIC WITH 2,6-ANTHRAQUINONE. *Chem. Lett.* **1983**, *12* (6), 905–908.
- (31) Cheung, C. W.; Buchwald, S. L. Mild and general palladium-catalyzed synthesis of methyl aryl ethers enabled by the use of a palladacycle precatalyst. *Org. Lett.* **2013**, *15* (15), 3998–4001.
- (32) Gámez-Valenzuela, S.; Ma, S.; Wei, Y.; Lu, Y.; Xu, C.; Feng, K.; Jeong, S. Y.; Wang, H.; Woo, H. Y.; Facchetti, A.; Guo, X. One-Step Synthesis of Cyanated Quinoxaline Regioisomers Enables n-Type Polymers with Record Ambient-Processed Thermoelectric Performance. *J. Am. Chem. Soc.* **2025**, *147* (42), 38851–38864.
- (33) Velusamy, A.; Yu, C.-H.; Afraj, S. N.; Lin, C.-C.; Lo, W.-Y.; Yeh, C.-J.; Wu, Y.-W.; Hsieh, H.-C.; Chen, J.; Lee, G.-H.; Tung, S.-H.; Liu, C.-L.; Chen, M.-C.; Facchetti, A. Thienoisindigo (TII)-based quinoidal small molecules for high-performance n-type organic field effect transistors. *Adv. Sci.* **2021**, *8* (1), 2002930.
- (34) Ashraf, R. S.; Kronemeijer, A. J.; James, D. I.; Sirringhaus, H.; McCulloch, I. A new thiophene substituted isindigo based copolymer for high performance ambipolar transistors. *Chem. Commun.* **2012**, *48* (33), 3939–3941.
- (35) Sui, Y.; Deng, Y.; Du, T.; Shi, Y.; Geng, Y. Design strategies of n-type conjugated polymers for organic thin-film transistors. *Mater. Chem. Front.* **2019**, *3* (10), 1932–1951.
- (36) Shi, Y.; Li, J.; Sun, H.; Li, Y.; Wang, Y.; Wu, Z.; Jeong, S. Y.; Woo, H. Y.; Fabiano, S.; Guo, X. Thiazole imide-based all-acceptor homopolymer with branched ethylene glycol side chains for organic thermoelectrics. *Angew. Chem., Int. Ed.* **2022**, *61* (51), No. e202214192.
- (37) Hasegawa, T.; Ashizawa, M.; Hiyoshi, J.; Kawauchi, S.; Mei, J.; Bao, Z.; Matsumoto, H. An ultra-narrow bandgap derived from thienoisindigo polymers: structural influence on reducing the bandgap and self-organization. *Polym. Chem.* **2016**, *7* (5), 1181–1190.
- (38) Chang, S. H.; Chiang, C.-H.; Kao, F.-S.; Tien, C.-L.; Wu, C.-G. Unraveling the enhanced electrical conductivity of PEDOT: PSS thin films for ITO-free organic photovoltaics. *IEEE Photonics J.* **2014**, *6* (4), 1–7.
- (39) Mryasov, O. N.; Freeman, A. J. Electronic band structure of indium tin oxide and criteria for transparent conducting behavior. *Phys. Rev. B* **2001**, *64* (23), 233111.
- (40) Aguilar del Valle, M. d. P.; Garrido, L. F.; Alonso-Huítón, J. C.; Terrones Pacheco, L. a.; Cruz-Manjarrez, H.; Reyes-Gasga, J.; Pérez-Martínez, A. L.; Rodríguez-Gómez, A. Design, growth, and characterization of crystalline copper oxide p-Type transparent semiconductive thin films with figures of merit suitable for their incorporation into translucent devices. *Cryst. Growth Des.* **2022**, *22* (4), 2168–2180.
- (41) Liu, J.; Shi, Y.; Dong, J.; Nugraha, M. I.; Qiu, X.; Su, M.; Chiechi, R. C.; Baran, D.; Portale, G.; Guo, X.; Koster, L. J. A. Overcoming Coulomb Interaction Improves Free-Charge Generation and Thermoelectric Properties for n-Doped Conjugated Polymers. *ACS Energy Lett.* **2019**, *4* (7), 1556–1564.
- (42) Pasveer, W. F.; Cottaar, J.; Tanase, C.; Coehoorn, R.; Bobbert, P. A.; Blom, P. W. M.; de Leeuw, D. M.; Michels, M. A. J. Unified Description of Charge-Carrier Mobilities in Disordered Semiconducting Polymers. *Phys. Rev. Lett.* **2005**, *94* (20), 206601.
- (43) Liu, J.; Qiu, L.; Portale, G.; Koopmans, M.; ten Brink, G.; Hummelen, J. C.; Koster, L. J. A. N-Type Organic Thermoelectrics: Improved Power Factor by Tailoring Host-Dopant Miscibility. *Adv. Mater.* **2017**, *29* (36), 1701641.
- (44) Deng, X.-Y.; Zhang, Z.; Lei, T. Integrated Materials Design and Process Engineering for n-Type Polymer Thermoelectrics. *JACS Au* **2024**, *4* (11), 4066–4083.
- (45) Feng, K.; Guo, H.; Wang, J.; Shi, Y.; Wu, Z.; Su, M.; Zhang, X.; Son, J. H.; Woo, H. Y.; Guo, X. Cyano-Functionalized Bithiophene Imide-Based n-Type Polymer Semiconductors: Synthesis, Structure-Property Correlations, and Thermoelectric Performance. *J. Am. Chem. Soc.* **2021**, *143* (3), 1539–1552.
- (46) Gao, Y.; Ke, Y.; Wang, T.; Shi, Y.; Wang, C.; Ding, S.; Wang, Y.; Deng, Y.; Hu, W.; Geng, Y. An n-type conjugated polymer with low crystallinity for high-performance organic thermoelectrics. *Angew. Chem., Int. Ed.* **2024**, *63* (20), No. e202402642.
- (47) Lu, Y.; Yu, Z.-D.; Zhang, R.-Z.; Yao, Z.-F.; You, H.-Y.; Jiang, L.; Un, H.-I.; Dong, B.-W.; Xiong, M.; Wang, J.-Y.; Pei, J. Rigid Coplanar Polymers for Stable n-Type Polymer Thermoelectrics. *Angew. Chem., Int. Ed.* **2019**, *58* (33), 11390–11394.
- (48) Han, J.; Jiang, Y.; Tiernan, E.; Ganley, C.; Song, Y.; Lee, T.; Chiu, A.; McGuiggan, P.; Adams, N.; Clancy, P.; Russell, T. P.; Hopkins, P. E.; Thon, S. M.; Tovar, J. D.; Katz, H. E. Blended Conjugated Host and Unconjugated Dopant Polymers Towards N-type All-Polymer Conductors and High-ZT Thermoelectrics. *Angew. Chem., Int. Ed.* **2023**, *62* (23), No. e202219313.
- (49) Du, T.; Liu, Y.; Wang, C.; Deng, Y.; Geng, Y. n-Type Conjugated Polymers Based on an Indandione-Terminated Quinoidal Building Block. *Macromolecules* **2022**, *55* (14), 5975–5984.
- (50) Wang, T.; Xu, C.; Deng, Y.; Geng, Y. Conjugated Polymers with an Oxindole-Terminated Quinoidal Unit: Energy Level Modulation Toward Air-Stable n-Doped Conductors. *Adv. Funct. Mater.* **2025**, *35* (2), 2412647.
- (51) Cisneros-Contreras, I. R.; Muñoz-Rosas, A. L.; Rodríguez-Gómez, A. Resolution improvement in Haacke's figure of merit for transparent conductive films. *Results Phys.* **2019**, *15*, 102695.
- (52) Massetti, M.; Jiao, F.; Ferguson, A. J.; Zhao, D.; Wijeratne, K.; Würger, A.; Blackburn, J. L.; Crispin, X.; Fabiano, S. Unconventional thermoelectric materials for energy harvesting and sensing applications. *Chem. Rev.* **2021**, *121* (20), 12465–12547.
- (53) Russ, B.; Gludell, A.; Urban, J. J.; Chabinyc, M. L.; Segalman, R. A. Organic thermoelectric materials for energy harvesting and temperature control. *Nat. Rev. Mater.* **2016**, *1* (10), 16050.
- (54) Du, T.; Liu, Y.; Deng, Y.; Geng, Y. Enhanced n-type thermoelectric performance of conjugated polymers based on an indandione-terminated quinoidal unit through comonomer optimization. *Chin. J. Chem.* **2023**, *41* (7), 776–782.
- (55) Sun, W.; Gámez-Valenzuela, S.; Zhang, X.; Lee, J.-W.; Zhong, Z.; Wang, P.; Ma, S.; Wang, H.; Kim, B. J.; Guo, X. A Simple Quinoid Building Block for Polymer Semiconductors with Tunable Polarity and High n-Type Thermoelectric Performance. *Angew. Chem., Int. Ed.* **2025**, *64* (28), No. e202501196.
- (56) Dazzi, A.; Prater, C. B. AFM-IR: Technology and Applications in Nanoscale Infrared Spectroscopy and Chemical Imaging. *Chem. Rev.* **2017**, *117* (7), 5146–5173.
- (57) Yang, C.-Y.; Ding, Y.-F.; Huang, D.; Wang, J.; Yao, Z.-F.; Huang, C.-X.; Lu, Y.; Un, H.-I.; Zhuang, F.-D.; Dou, J.-H.; Di, C.-a.; Zhu, D.; Wang, J.-Y.; Lei, T.; Pei, J. A thermally activated and highly miscible dopant for n-type organic thermoelectrics. *Nat. Commun.* **2020**, *11* (1), 3292.